

Question D.3 — Stratification by Amplitude and the Aggregation Artefact

Every coefficient here is Kendall’s τ_b with censored runs imputed to the last rank, justified in Q D.2. Every confidence interval (CI) is a bootstrap interval over the seeds of the stratum, 2000 resamples, quantiles 2.5% and 97.5%. Values are produced by `scripts/chiffres_QD.py` from the campaign of 2000 runs (20 amplitudes $\times$ 100 seeds, $H = 2000$, $M = 10$ trees in the adaptive random forest, ARF).

1 Why Stratify

Δe drives both sides of every correlation of interest: it shortens τ_{ARF} and it raises the evidence available to the detector. A coefficient computed on the pooled 2000 runs therefore measures the common dependence on Δe rather than any association between the two quantities at a fixed operating point. Stratification removes that common cause. Each stratum holds 100 seeds, and the amplitude is constant inside it by construction.

2 The Aggregation Collapse

Comparator	pooled τ_b	median stratified τ_b (18 amplitudes)	ratio
$\tau_{\text{swap}}(10\%)$	+1.000000	+1.000000	1.00
$\tau_{\text{swap}}(25\%)$	+0.644626	+0.493224	1.31
$\tau_{\text{swap}}(50\%)$	+0.556494	+0.336523	1.65
$\tau_{\text{swap}}(75\%)$	+0.461827	+0.097934	4.72
$A(H)$	+0.396512	+0.052559	7.54
$S_{\text{max}}(H)$	+0.314673	+0.212865	1.48

Table 1: Pooled against stratified coefficients. The first row is tautological, see Section 3.

For $A(H)$ the pooled coefficient is +0.3965 against a median stratified coefficient of +0.0526, a factor of 7.5. This is not a strong association weakened by stratification: the stratified coefficient is indistinguishable from zero under this protocol at every one of the 18 amplitudes. The pooled coefficient manufactures a link rather than confirming a weak one. Mechanically it is Simpson’s paradox on Δe : within each stratum no association is discernible, and across strata both quantities move with the amplitude, so pooling recovers the amplitude and reports it as an association.

The effect is strongest exactly where the within-stratum association is weakest: ratio 7.54 for $A(H)$ and 4.72 for $\tau_{\text{swap}}(75\%)$, but only 1.48 for $S_{\text{max}}(H)$, which does carry a genuine within-stratum signal. That is the diagnostic one should expect, and it is why no pooled coefficient is quoted as a result in this work.

3 A Tautological Row, Marked as Such

At $M = 10$ a quota of 10% requires exactly one tree, so $\tau_{\text{swap}}(10\%) = \tau_{\text{swap}}(1/M) = \tau_{\text{ARF}}$ by definition (Q C.1a). Its τ_b against τ_{ARF} is 1.000000 at all 20 amplitudes with confidence interval $[1.000; 1.000]$, by identity rather than by performance. The row is kept in Table 1 and hatched in Figure 1 rather than deleted, so the reader can check that the instrumentation reproduces an identity it should reproduce. It carries no evidence about anything else. The precedent is the treatment of $S_{\text{max}}(H)$ under test B2

elsewhere in this work, whose area under the ROC curve (AUC) of 1.000 is tautological in the same sense.

A related caution applies to the other quotas. At $M = 10$ neither $q = 25\%$ nor $q = 75\%$ is exactly attainable: 25% requires 3 trees, that is 30%, and 75% requires 8, that is 80%. The labels are kept for continuity with Q C.1, which uses the same convention, but they name the attainable quota above the nominal one.

4 Coefficients by Amplitude

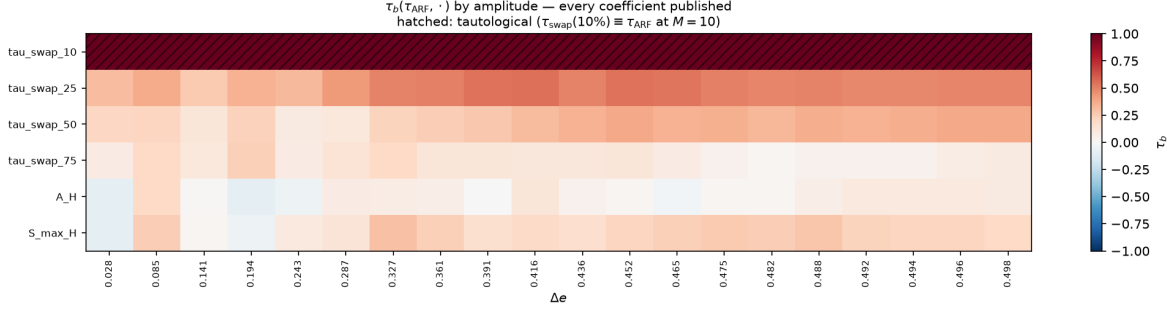


Figure 1: $\tau_b(\tau_{ARF}, \cdot)$ for every comparator at every amplitude. All 120 coefficients are published, with no post hoc selection. The hatched row is tautological.

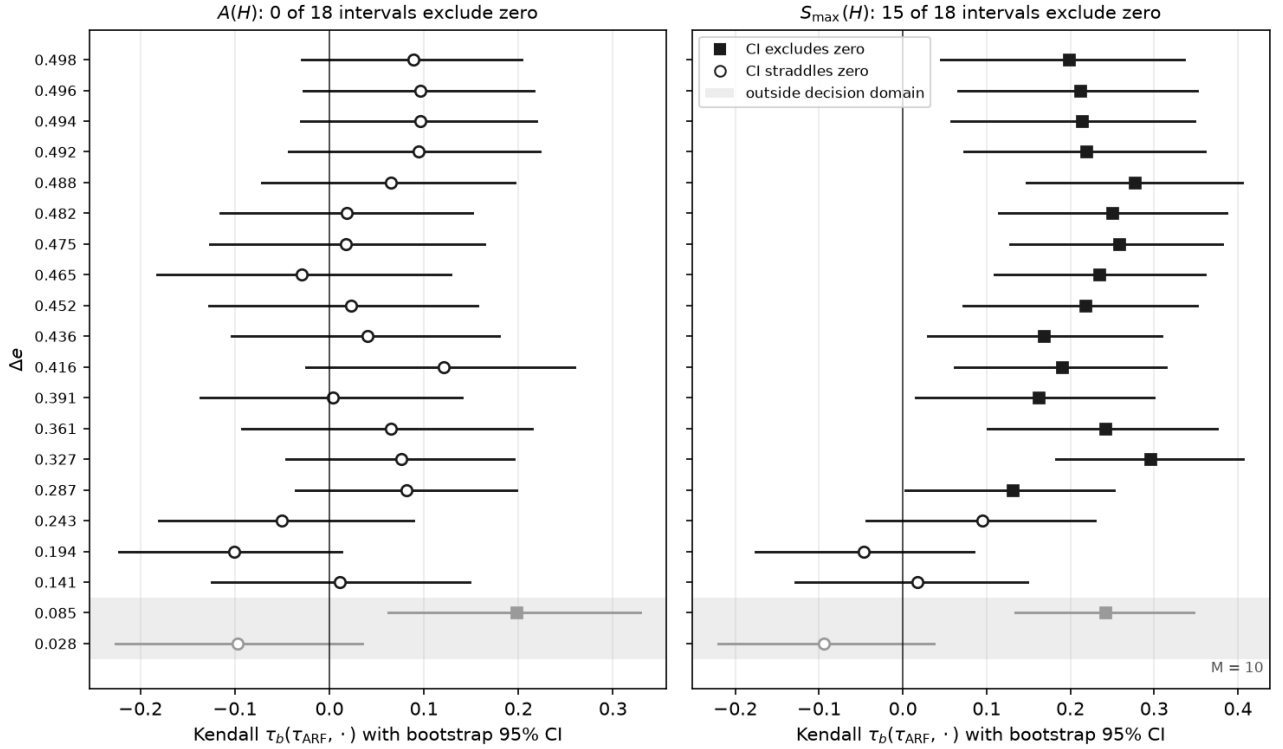


Figure 2: Forest plot of $\tau_b(\tau_{ARF}, \cdot)$ with its bootstrap 95% confidence interval, for $A(H)$ (left) and $S_{\max}(H)$ (right), one row per amplitude; filled squares mark intervals that exclude zero, hollow circles those that straddle it, and the two shaded rows lie outside the decision domain. 15 of 18 intervals exclude zero for $S_{\max}(H)$, none for $A(H)$; rule of the interval, see Q D.4. The 15 are the 15 consecutive amplitudes from $\Delta e = 0.287$ upward. Table QCD_correlations_stratified_full.

5 The Central Result of Q D

The two quantities in Table 2 play different roles, and conflating them changes the answer.

Δe	$A(H)$			$S_{\max}(H)$		
	τ_b	CI low	CI high	τ_b	CI low	CI high
0.141	+0.0111	-0.1249	+0.1485	+0.0174	-0.1288	+0.1487
0.194	-0.1009	-0.2228	+0.0127	-0.0470	-0.1759	+0.0846
0.243	-0.0504	-0.1802	+0.0887	+0.0952	-0.0432	+0.2294
0.287	+0.0815	-0.0357	+0.1978	+0.1308	+0.0027	+0.2519
0.327	+0.0762	-0.0460	+0.1954	+0.2954	+0.1832	+0.4063
0.361	+0.0651	-0.0924	+0.2146	+0.2413	+0.1009	+0.3748
0.391	+0.0037	-0.1366	+0.1398	+0.1623	+0.0154	+0.3002
0.416	+0.1212	-0.0245	+0.2595	+0.1904	+0.0621	+0.3141
0.436	+0.0402	-0.1037	+0.1796	+0.1682	+0.0295	+0.3088
0.452	+0.0230	-0.1277	+0.1570	+0.2177	+0.0722	+0.3511
0.465	-0.0296	-0.1820	+0.1279	+0.2347	+0.1095	+0.3611
0.475	+0.0171	-0.1260	+0.1642	+0.2584	+0.1282	+0.3817
0.482	+0.0179	-0.1151	+0.1509	+0.2496	+0.1141	+0.3861
0.488	+0.0650	-0.0710	+0.1963	+0.2764	+0.1478	+0.4055
0.492	+0.0945	-0.0432	+0.2226	+0.2193	+0.0736	+0.3608
0.494	+0.0964	-0.0301	+0.2188	+0.2135	+0.0577	+0.3478
0.496	+0.0966	-0.0273	+0.2164	+0.2122	+0.0659	+0.3518
0.498	+0.0890	-0.0293	+0.2038	+0.1985	+0.0457	+0.3359

Table 2: $\tau_b(\tau_{\text{ARF}}, \cdot)$ with bootstrap confidence intervals over the seeds, on the 18 amplitudes of the decision domain.

$A(H)$ is the accumulated excess error over the horizon. Its association with τ_{ARF} is indistinguishable from zero under this protocol at all 18 amplitudes: median +0.0526, every confidence interval straddling zero.

$S_{\max}(H)$ is the maximum of the path of the cumulative-sum (CUSUM) detector of Page, $S_t = \max(0, S_{t-1} + e_t - \hat{p}_0 - \delta_P)$. Its association is discernible at 15 of the 18 amplitudes, with a median of +0.2129, and the confidence interval lies entirely above zero at every amplitude from $\Delta e = 0.287$ upwards. The count is that of the confidence-interval rule adopted in Q D.4; the threshold rule of the **discernable** column gives 14, differing on the single cell $\Delta e = 0.287$.

The distinction matters because it is $S_{\max}(H)$ that decides the alarm, not $A(H)$. The alarm is by definition the crossing of λ by $S_{\max}$. A claim that τ_{ARF} carries no information about the evidence offered to the detector is attached to the wrong object if it rests on $A(H)$ alone.

Why $A(H)$ shows nothing. $A(H)$ sums $e_t - \hat{p}_0$ over 2000 steps, so an error on the estimated baseline is multiplied by 2000. Within a stratum, $\tau_b(\hat{p}_0, A(H))$ has a median of -0.4741 over the 18 amplitudes, and the baseline accounts for a median share of 0.51 of the run-to-run variance of $A(H)$, up to 0.88. Over the short window $w = \text{round}(3 \times 18.5 / \Delta e)$ the same coefficient falls to -0.0639, and an association with τ_{ARF} appears: median +0.2313, with a bootstrap interval excluding zero at 13 of the 18 amplitudes (exploratory, reproduced by `chiffres_QD.py -sous D3w`). The answer, stated sharply:

τ_{ARF} shows **no discernible association** with the full-horizon excess error $A(H)$ at any of the 18 amplitudes, but about half of the variance of $A(H)$ is baseline noise multiplied by H ; over the transient window alone the association is weak but real. It carries a **weak but real** association with $S_{\max}(H)$, the quantity that determines whether the alarm fires: median $\tau_b = +0.2129$, discernible at 15 of 18 amplitudes, that is every amplitude from $\Delta e = 0.287$ upwards without exception. Below $\Delta e = 0.287$ the association is not distinguishable from zero either.

Read with Q D.2, ties on τ_{ARF} are too rare to move any of these verdicts: they displace coefficients towards zero, but only 0.2% to 4.5% of the pairs of a stratum are tied.

6 Reservations

Multiplicity. The decision domain contains 108 cells, of which 68 are significant at the 5% level with no correction for multiplicity. The signal for $S_{\max}(H)$ above $\Delta e = 0.287$ is far too systematic to be a multiplicity artefact, running to 15 consecutive amplitudes in the same direction (Figure 2), but individual cells whose $|\tau_b|$ is near the detectability threshold are not quoted on their own.

Detectability threshold. The threshold used by the `discernable` column is 0.13295637, a single value across the table. It is quoted exactly rather than rounded, since a rounded threshold would not reproduce the column. Q D.4 examines the three cells where the two rules disagree and fixes which one governs.

Scope. Everything above is measured at $M = 10$ on one campaign. The two amplitudes below $\Delta e = 0.10$ fall outside the decision domain and are excluded from the medians quoted here, though they appear in Figure 1.